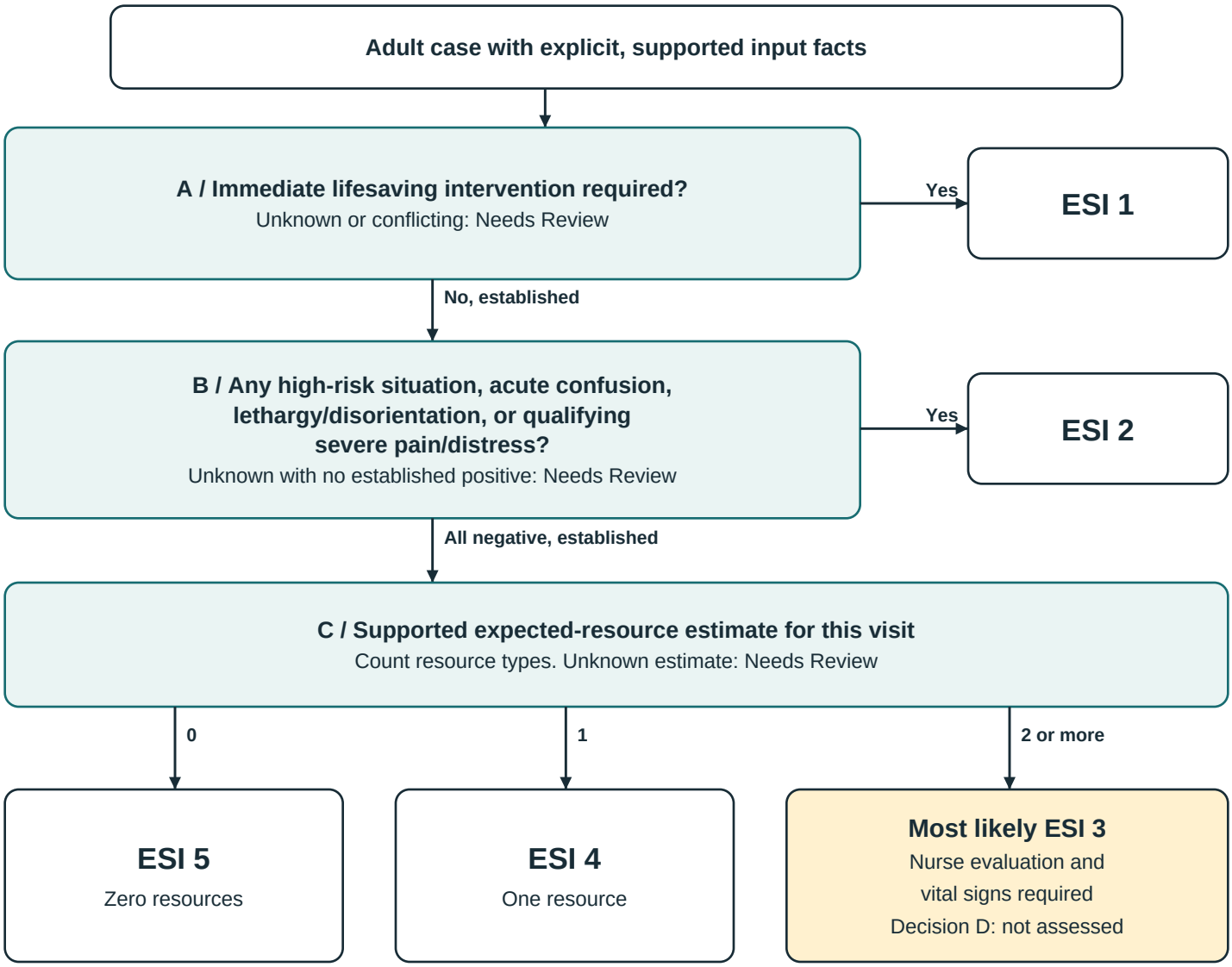
ESI rules with a provisional level 3

For nurse review of the decision procedure. The amber branch is a proposed study adaptation; approval and patient-input mappings remain pending.



A/B/C uncertainty -> Needs Review, no numeric level. It never defaults to 3.

In this no-vitals study, every prediction of 3 remains provisional.

How to read this diagram

An explicit positive at A takes precedence. After A is negative, one established positive B criterion is sufficient for level 2. Proceeding to C requires all B criteria negative. Normal execution also requires documented rule approval.

The phrase "most likely" is a display label, not a measured probability. It does not certify normal vital signs or that waiting is safe. The system records a numeric prediction of 3 separately from the pending nurse assessment.

Source: Gilboy et al., ESI v4 Implementation Handbook (2012 Edition), Chapter 2, pp. 8-14; Chapter 4, pp. 29-31; Chapter 5, pp. 35-37. The standard clinical D path is shown in the notes on page 2.

Clinical rules and boundaries

The software stores rule IDs, decision point, required fields, evidence, source, approval status and rule version. The rule sequence is deterministic.

ESI4-A | Immediate lifesaving intervention

An established need for immediate airway, breathing, circulatory or qualifying emergency medication intervention supports ESI 1. Handbook examples include apnea, pulselessness, acute unresponsiveness and severe respiratory compromise requiring lifesaving support. IV access, oxygen, ECG, monitoring or tests alone do not establish this need. Being able to type is not an exclusion.

Chapter 2, pp. 8-10; Table 2-1.

ESI4-B-HIGH-RISK | High-risk situation

After A is negative, an established high-risk situation supports ESI 2. Examples include active chest pain suspicious for ACS, stroke signs, healthcare-worker needle stick, fever during chemotherapy, and suicidal/homicidal presentation. These require context; raw symptom keywords do not establish a diagnosis or risk assessment.

Chapter 2, pp. 11-12; Chapter 3.

ESI4-B-MENTAL-STATUS | Acute change

After A is negative, new confusion, lethargy or disorientation supports ESI 2. Distinguish an acute change from the patient's baseline. Ordinary fatigue and chronic confusion alone do not satisfy this criterion.

Chapter 2, p. 12.

ESI4-B-PAIN-DISTRESS | Qualifying severe pain/distress

After A is negative, an assessment that severe pain or physiological/psychological distress requires high-priority care supports ESI 2. A reported score of 7/10 or higher prompts consideration; it is not an automatic ESI 2 rule.

Chapter 2, pp. 12-13; Chapter 3.

ESI4-C | Expected resources

After A and all B criteria are negative, 0 expected resource types supports ESI 5, 1 supports ESI 4, and at least 2 proceeds to D. A supported estimate of the current visit is required. Existing tests or catheters are not automatic resource estimates.

Chapter 4, pp. 29-31; Table 4-1.

ESI4-D-PROVISIONAL | Proposed study adaptation

With supported negative A/B decisions and at least 2 expected resources, return provisional numeric 3 with nurse evaluation and vital signs required. If A/B/C remains unclear, return Needs Review. This defers D; it does not complete the handbook's ESI 3 assessment.

Adaptation informed by Chapter 2, p. 14; Chapter 5, pp. 35-37.

Separate clinical D path: adult HR >100/min, RR >20/min or oxygen saturation <92% prompts consideration of ESI 2 with clinical judgment. Values outside the danger zone support ESI 3 after A/B/C. Missing measurements remain unresolved. This measured-vitals path is outside the patient-only experiment.

What the nurse is being asked to review

Review the rules above and the proposed deferral at D. Existing validated case labels can be retained. Rule approval and the link from patient reports to these required facts are separate review decisions.

The current input gap

The chatbot extracts verbatim excerpts from 15 patient fields. It does not yet establish a lifesaving-intervention assessment, each B assessment, or an expected-resource estimate. The preview supplies these facts explicitly in fictional examples. Those examples demonstrate software behavior; they cannot supply missing facts in the final study.

Required decision	Review issue for the existing patient inputs
A: intervention need	Which explicit reported conditions can support a positive or negative assessment? Ability to type and absence of a keyword are insufficient exclusions.
B: risk, mental status, pain/distress	Specify the context, acute change and qualifying distress evidence. Preserve unknown or contradictory reports. A pain score alone is insufficient.
C: resource estimate	Can a general rule establish 0, 1, or at least 2 expected resources from the permitted facts? Record the exact conditions and rationale. If the inputs cannot support it, retain Needs Review.
D: provisional 3	Accept, revise or reject numeric 3 with D explicitly unassessed, only after supported A/B/C decisions. This does not establish clinical stability.

Resource reference for discussion

Count expected resource types: labs, ECG, imaging, IV hydration, IV/IM/nebulized medication, specialty consultation and procedures. Simple procedure = 1; complex procedure such as conscious sedation = 2. CBC + urinalysis = one lab resource; CBC + X-ray = two types.

History/examination, point-of-care tests, oral medication, prescription refills, tetanus immunization, simple dressings and crutches/splints/slings are not counted ESI resources. A past/requested test list does not establish the expected complete workup. An existing catheter does not establish that insertion is needed.

Review record

Decision: Approve as specified / Revise / Unsupported with these inputs
Rule ID(s): _____
Conditions, exclusions and revisions: _____
_____
Reviewer name/role: _____ Date: _____
Review-record ID/signature: _____

Approval must identify the exact rule definition/version. The provisional D adaptation needs its own review. Generated text, software fixtures and an unsigned diagram are not approval records.

How the outcomes count in final testing

Keep processing status, predicted level and pending clinical assessment in separate columns. Apply the frozen scoring definitions to both methods.

Recorded outcome	Treatment in the comparison
Provisional numeric ESI 1-5	Use the numeric prediction. Provisional 3 is correct only when the reference is 3. Report its frequency and errors separately; nurse evaluation pending is not itself an abstention.
Needs Review; priority = null	Count as an abstention. No sixth ESI class, no imputed 3, and no overtriage/undertriage classification. Keep it in attempted-input denominators.
Technical failure	Report separately from Needs Review. Retain it in attempted-input denominators.
Draft preview / unapproved demonstration	Exclude from final study runs. Candidate priorities shown in the review command are not accepted study predictions.

Metrics for each run

Conditional comparison: compute accuracy, macro-F1, confusion matrices and error direction on the common paired set where both methods returned numerical levels. Report its size and coverage.

Coverage: numerical predictions / all attempted inputs, separately for A and B. Also report common-pair coverage.
Needs Review rate: abstentions / attempted inputs. Break down by reference ESI level, language and reason.
Completion-adjusted accuracy: correct numerical predictions / attempted inputs. Abstentions stay in this denominator.
Technical-failure rate: failed inputs / attempted inputs, reported separately.

Consistency and repeatability: under the paper's definition, require valid identical numeric levels. A case group with Needs Review is not successfully consistent. Describe agreement between abstention statuses separately if useful.

Worked illustration - not study results

If A predicts all 100 inputs and B predicts 80, with 20 Needs Review outcomes and 60 correct B predictions: conditional accuracy = 60/80 = 75%; B coverage = 80%; Needs Review = 20%; completion-adjusted accuracy = 60/100 = 60%. If both methods abstain on different inputs, compute the common paired set explicitly.

Three repetitions

Report each run and the arithmetic mean. Preserve all case/version/run IDs and canonical-case grouping. Planned final evaluation: 600 input versions x 3 runs = 1,800 paired records from 120 underlying cases. If no valid pairs exist, conditional comparison is not estimable. Do not select the best run or replace a frozen prediction with a later nurse result.

Source links and software review

AHRQ ESI v4 handbook | Sharma et al. (2026): provisional pre-vital phase
The article supports studying provisional prediction; it does not approve this deterministic adaptation. Statistical handling follows the thesis's Section 3.13.

Run locally: php artisan esi:preview --no-interaction
Rule version: esi-v4-adult-1.1.0-provisional-d+4f53cda18c2baa0c
Method A remains unchanged. Clinical approvals remain pending.